

Chromatic quasisymmetric functions for signed graphs

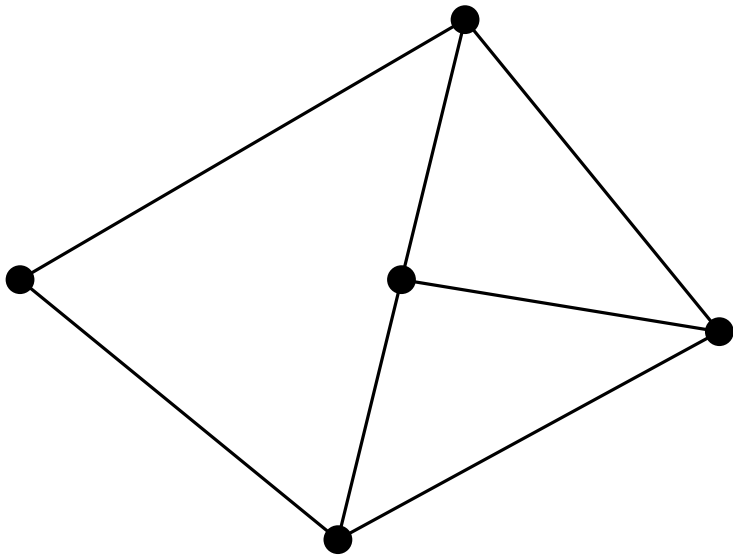
Raquel Melgar (joint work with Jean-Christophe Aval)

**Séminaire de l'équipe Combinatoire et Interactions du
LaBRI (June 23th, 2025)**

I. Introduction and preliminaries

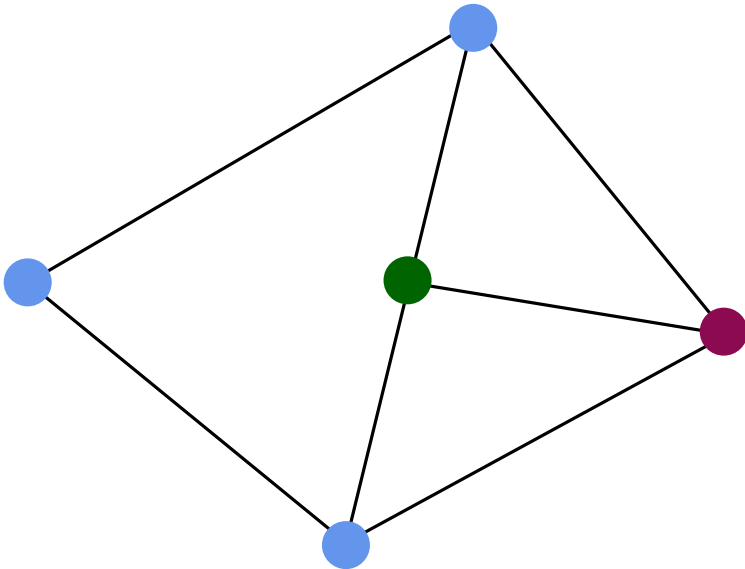
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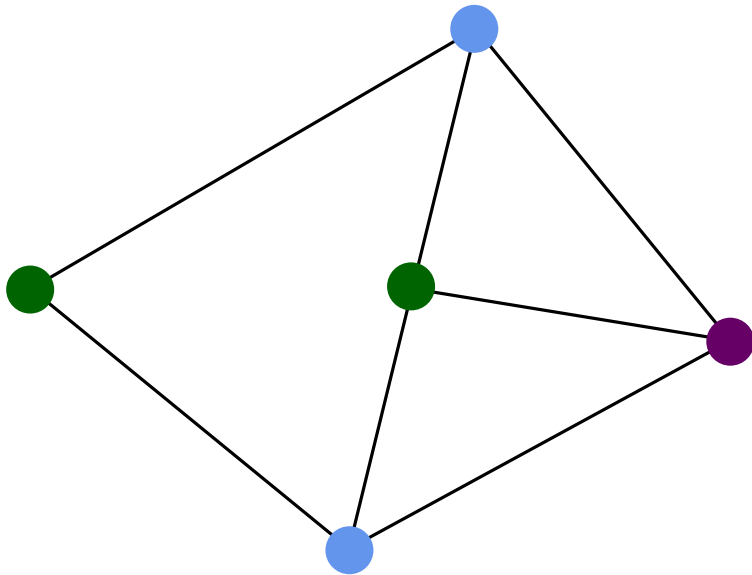
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This is **not** a proper coloring of G

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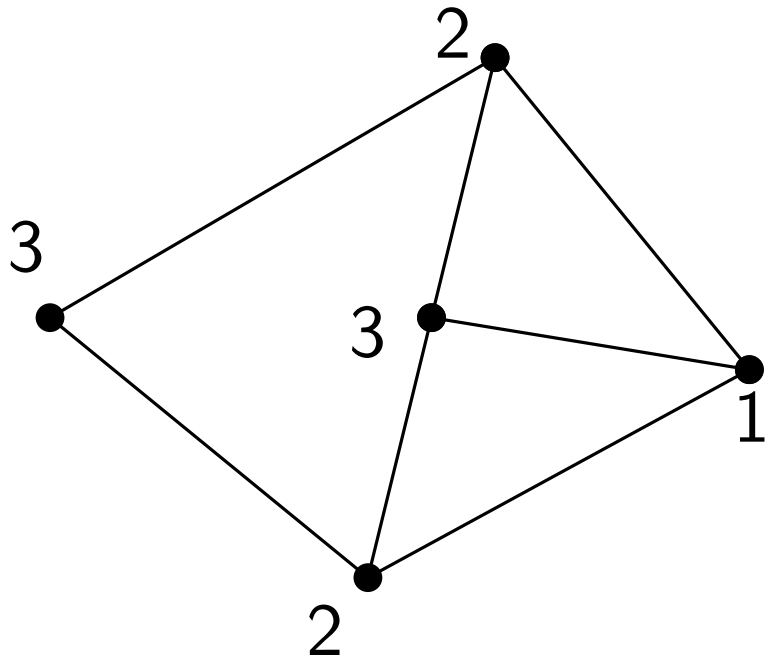
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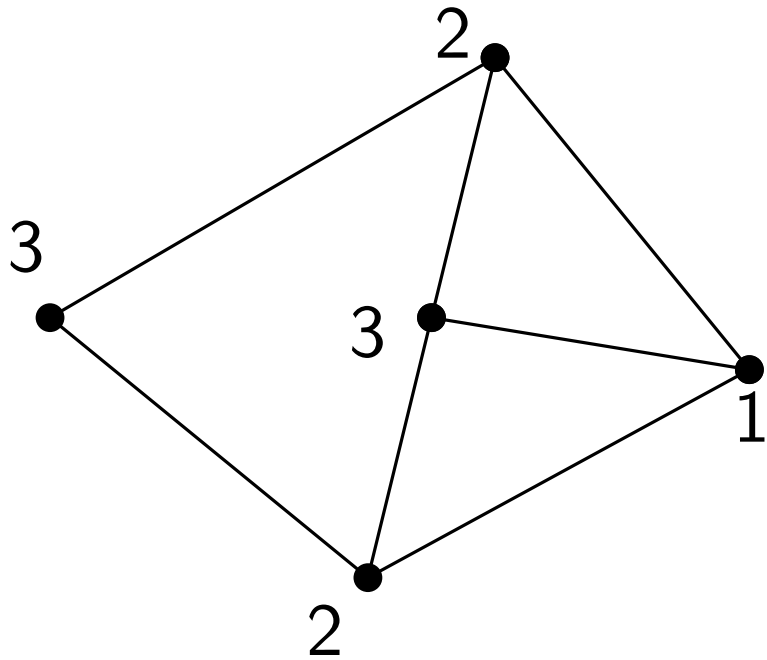
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A proper coloring of a graph G is a map $\kappa : V \rightarrow \mathbb{N}$ verifying $\kappa(i) \neq \kappa(j)$ if $\{i, j\} \in E$

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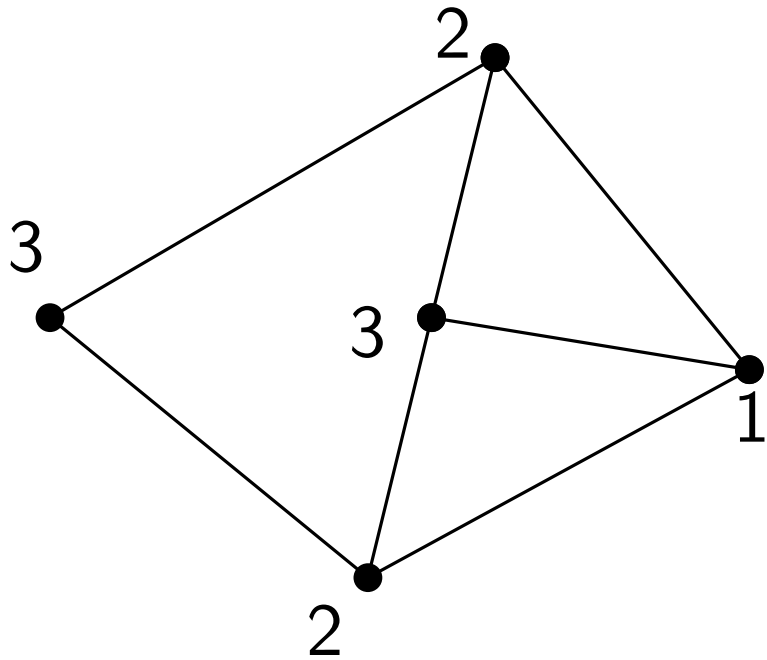


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- $\chi_G(k) :=$ number of (proper) colorings of G with k colors

The algebra of symmetric functions

The algebra of symmetric functions

Given a set of variables $x = (x_1, x_2, \dots)$ we say a formal power series $f \in \mathbb{Q}[[x_1, x_2, \dots]]$ is a symmetric function if it satisfies:

- f has bounded degree
- $f(x_1, x_2, \dots) = f(x_{\sigma(1)}, x_{\sigma(2)}, \dots) \quad \forall \sigma \in \mathfrak{S}$

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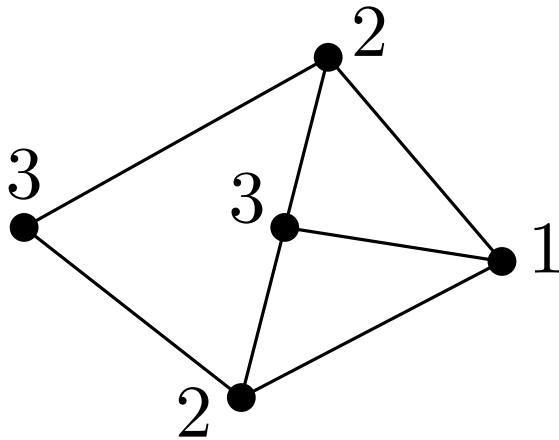
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The bases of Sym^d are indexed by the partitions of d

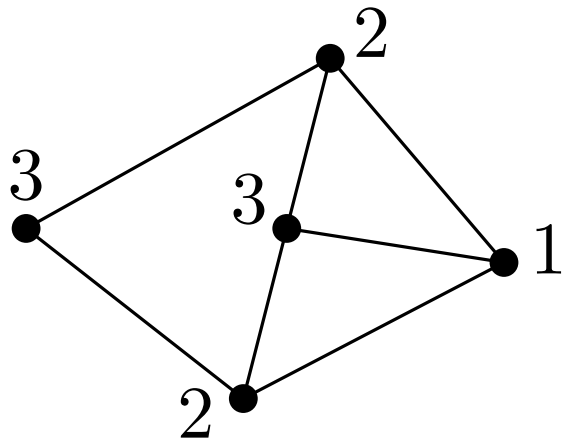
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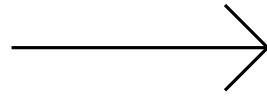


coloring κ

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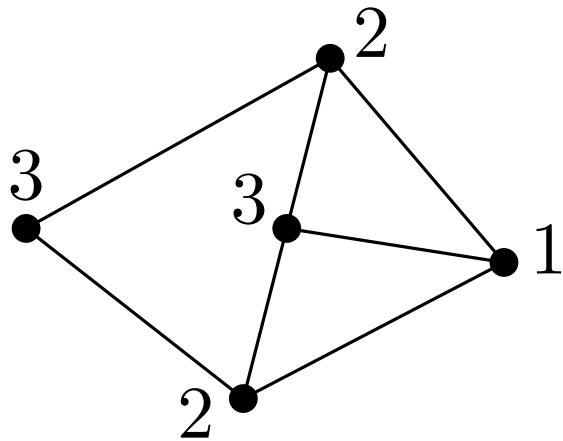


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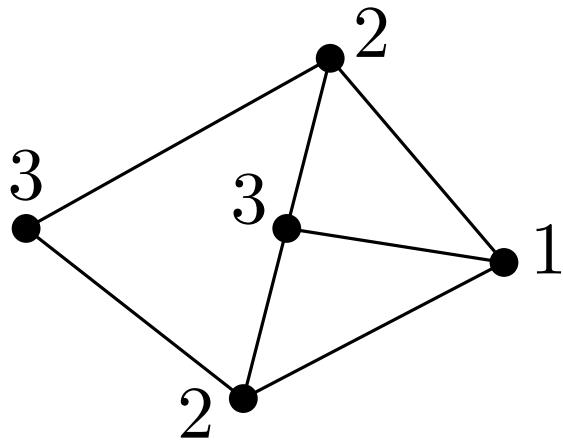
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In 1995 Stanley defined the *chromatic symmetric function* of a graph G as:

$$X_G(x_1, x_2, \dots) = \sum_{\kappa \text{ proper}} x^\kappa \in \text{Sym}^{|V|}$$

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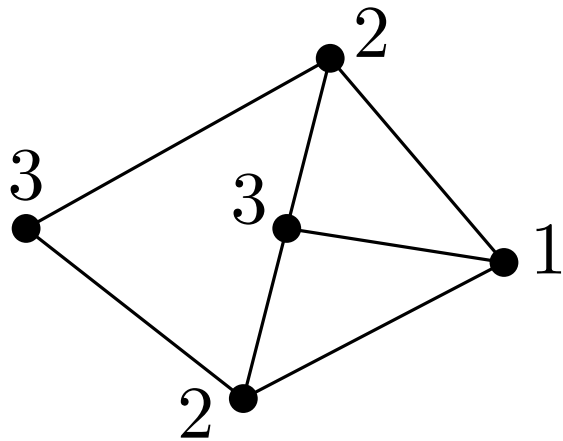
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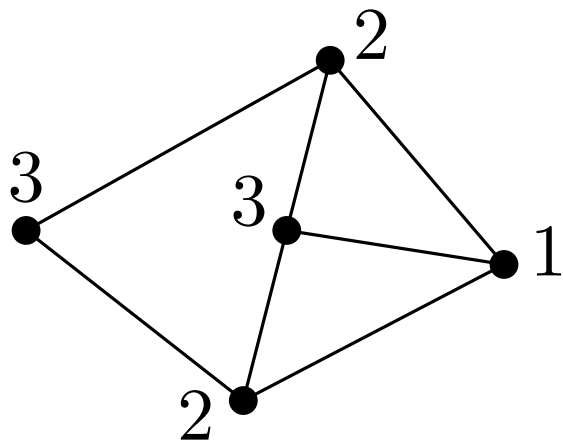
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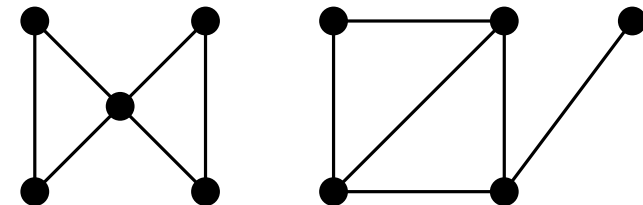
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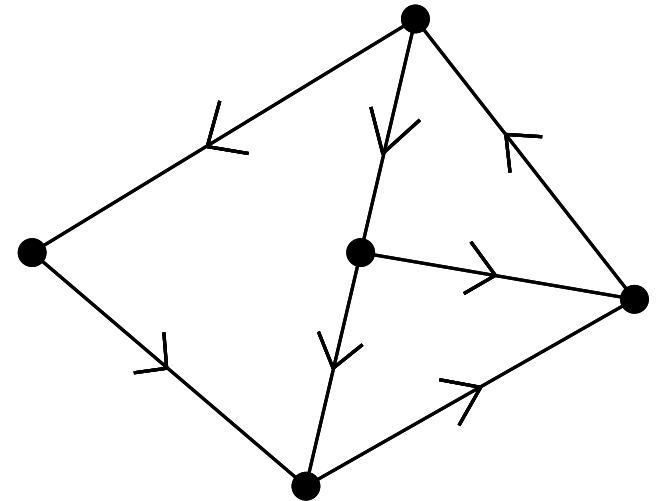
Does X_G distinguish non isomorphic graphs? **NO!**

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Ellzey's quasisymmetric invariant

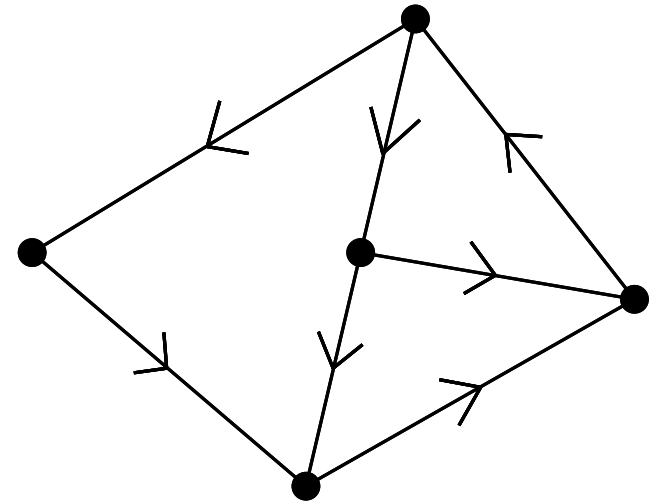
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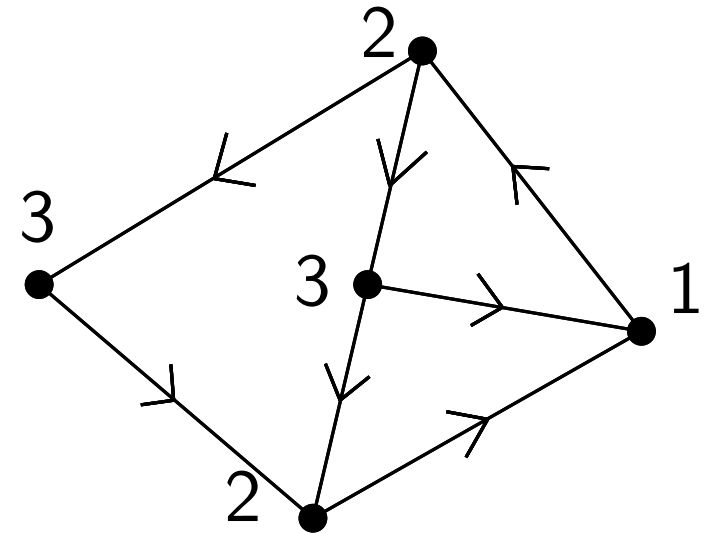
Given a directed graph $\vec{G}$ and a coloring κ of the underlying graph G we define an *ascent* of κ to be a directed edge $(i, j) \in E$ verifying $\kappa(i) < \kappa(j)$



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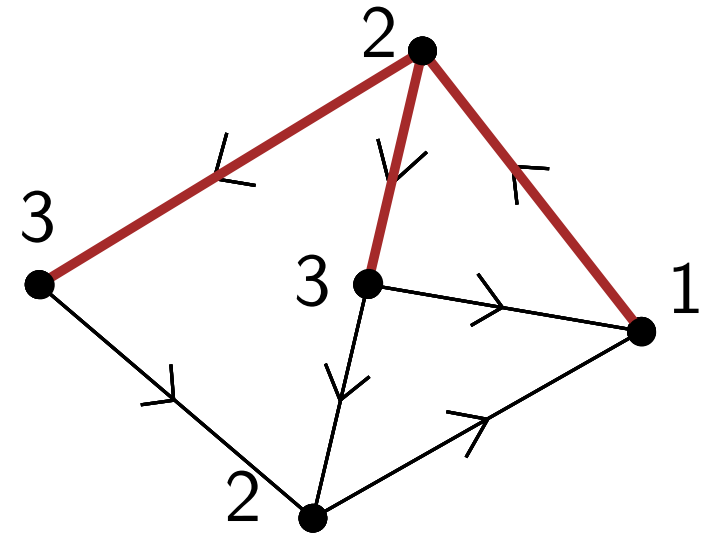
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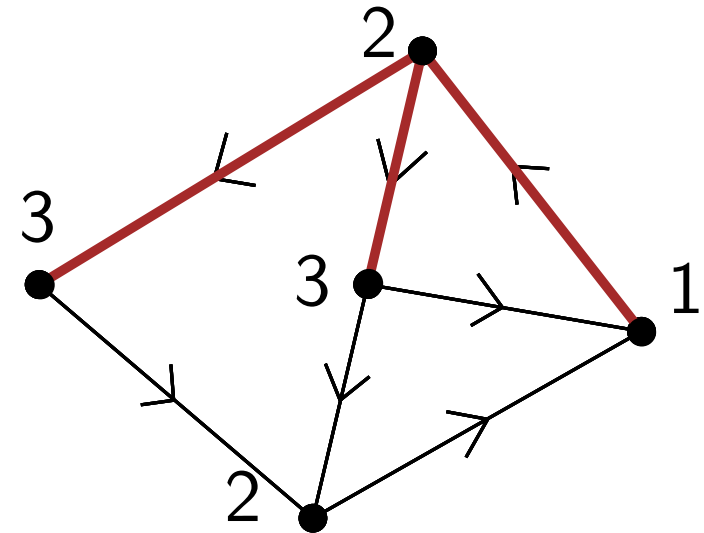


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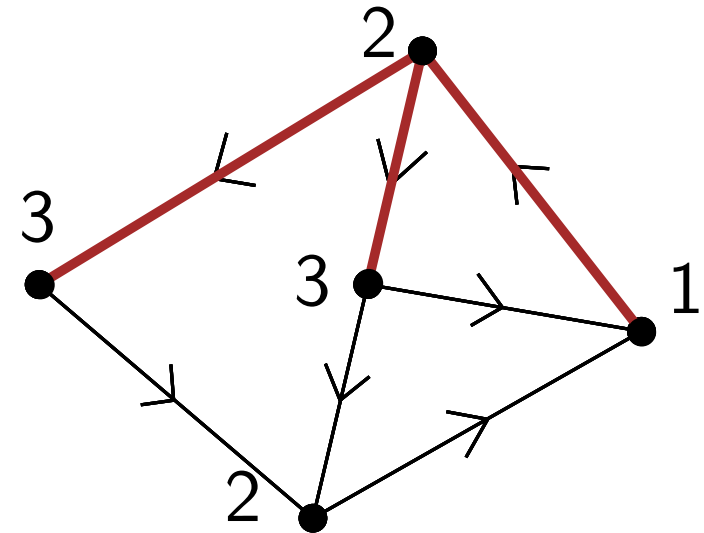


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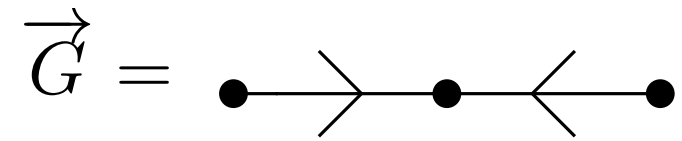


Then the chromatic quasisymmetric function of a digraph is:

$$X_G(x; t) := \sum_{\kappa} x^{\kappa} \cdot t^{asc(\kappa)}$$

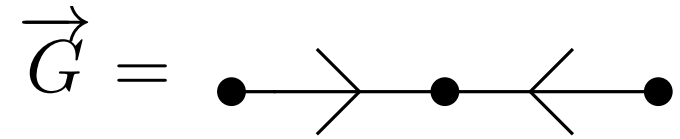
An example

$$X_{\vec{G}} = \sum_{\kappa} x^{\kappa} t^{asc(\kappa)}$$



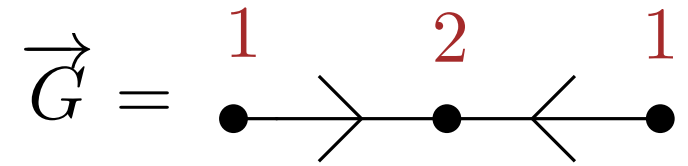
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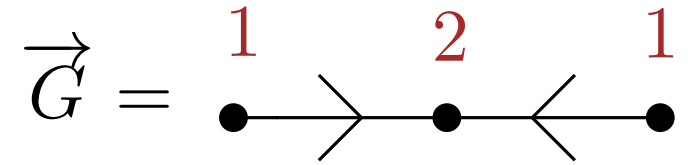
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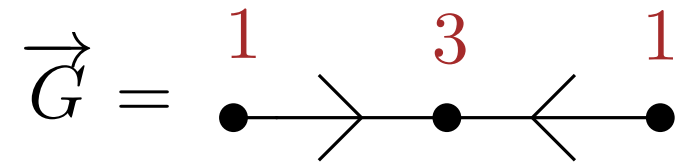
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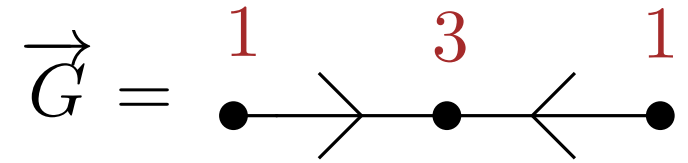
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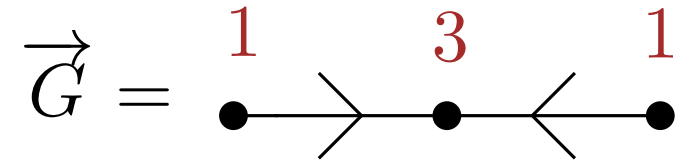
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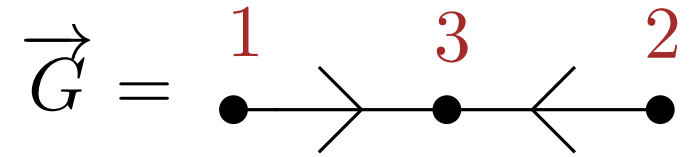
$$= t^2 (x_1^2 x_2 + x_1^2 x_3 + x_2^2 x_3 + \dots)$$



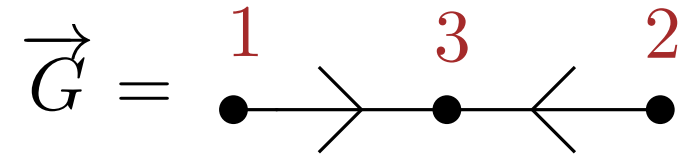
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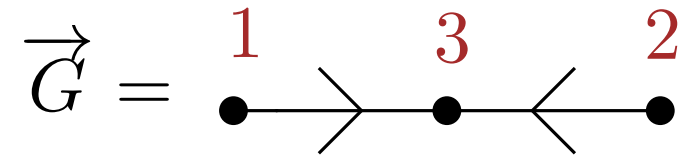
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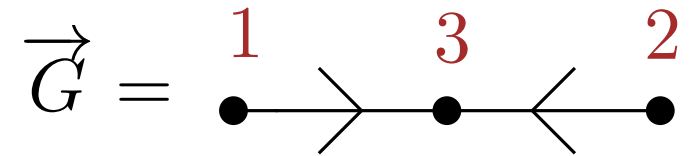
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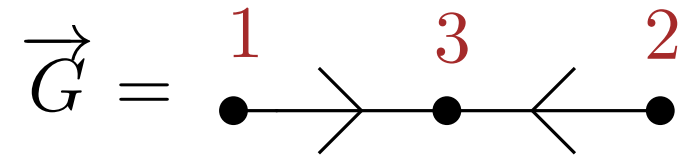


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Quasisymmetric functions

$$X_G(x; t) = \sum_{\kappa} x^{\kappa} \cdot t^{\text{asc}(\kappa)} \in QSym[t]$$

Intuition on quasisymmetric functions:

$$x_1 x_2^2 x_3$$

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For every composition

$$c = (c_1, \dots, c_r) \quad \rightarrow \quad M_c := \sum_{i_1 < i_2 < \cdots < i_r} x_{i_1}^{c_1} x_{i_2}^{c_2} \cdots x_{i_r}^{c_r}$$

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We can define

$$QSym := \text{span}\{M_c : c \text{ composition}\}$$

Quasisymmetric functions

We say a formal power series $f \in \mathbb{Q}[[x_1, x_2, \dots]]$ is a *quasisymmetric function* if:

- f has bounded degree
- f is "shifting invariant", this is: for any positive integer sequence $(c_1, c_2, \dots, c_r)$ we have

$$[x_{i_1}^{c_1} x_{i_2}^{c_2} \dots x_{i_r}^{c_r}] f = [x_{j_1}^{c_1} x_{j_2}^{c_2} \dots x_{j_r}^{c_r}] f$$

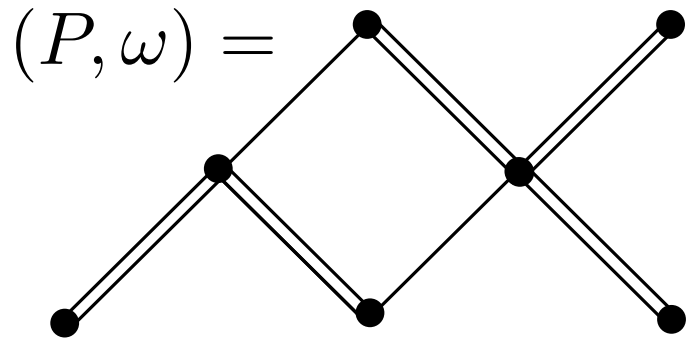
whenever $i_1 < i_2 < \dots < i_r$ and $j_1 < j_2 < \dots < j_r$.

Example and the fundamental basis

The enumerators of (P, ω) -partitions are quasisymmetric functions

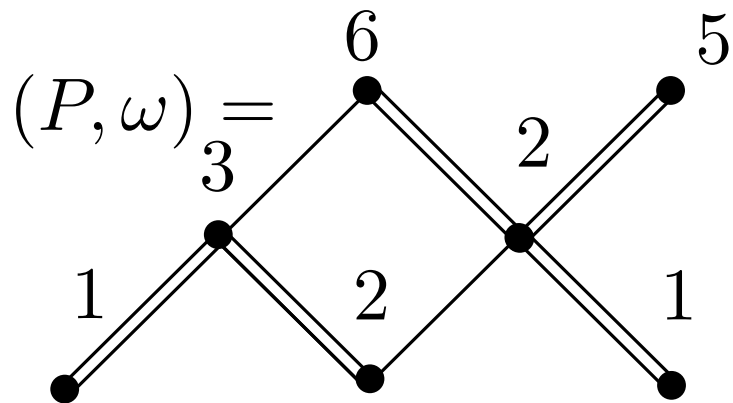
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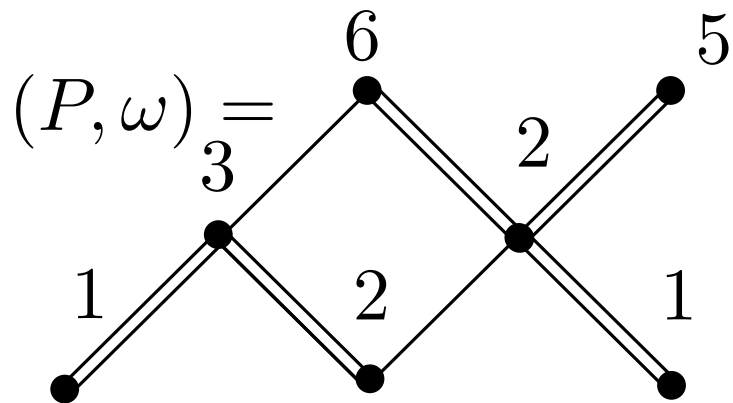
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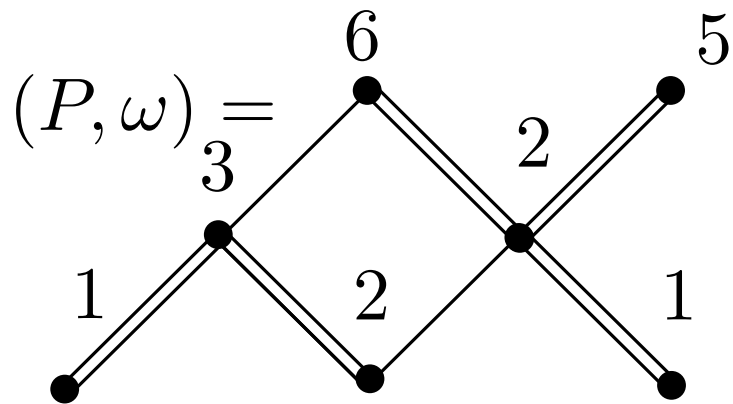
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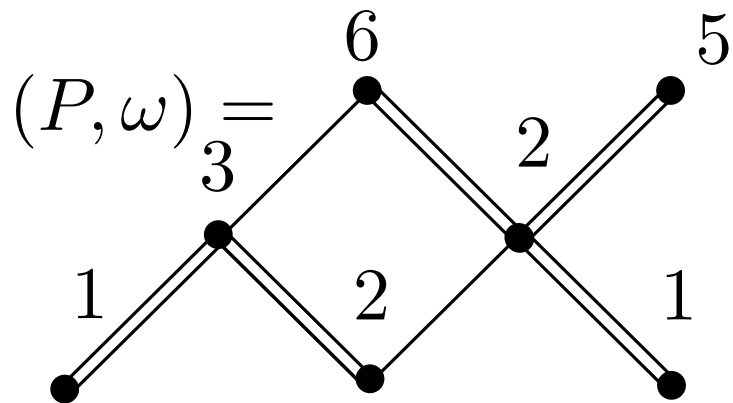
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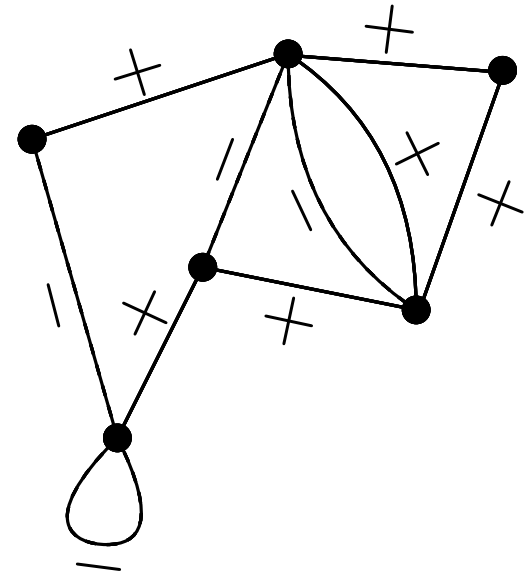
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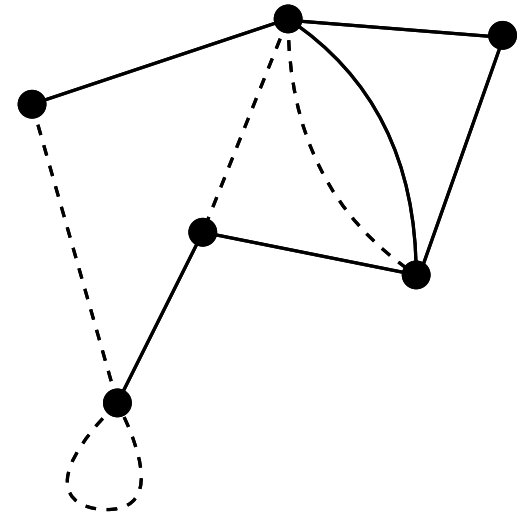
$$\Gamma\left(\begin{array}{c} \bullet \\ \parallel \\ \bullet \\ | \\ \bullet \end{array}\right) = \sum_{0 < i_1 \leq i_2 < i_3} x_{i_1} x_{i_2} x_{i_3}$$

is an element of the
fundamental basis

Signed graphs

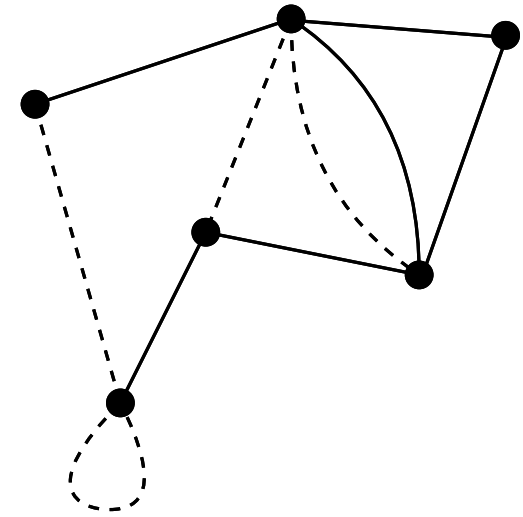


Signed graphs



Signed graphs

What is a proper coloring of a signed graph?



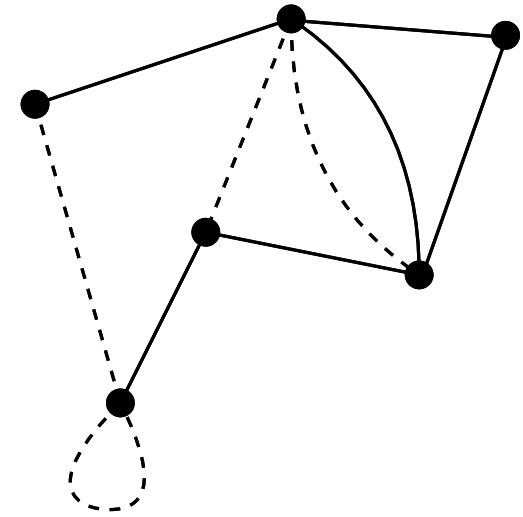
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A proper coloring of a signed graph

$\Sigma = (V, E^+, E^-)$ is a map $\kappa : V \rightarrow \mathbb{Z}$ verifying:

- $\kappa(i) \neq \kappa(j)$ if $\{i, j\} \in E^+$
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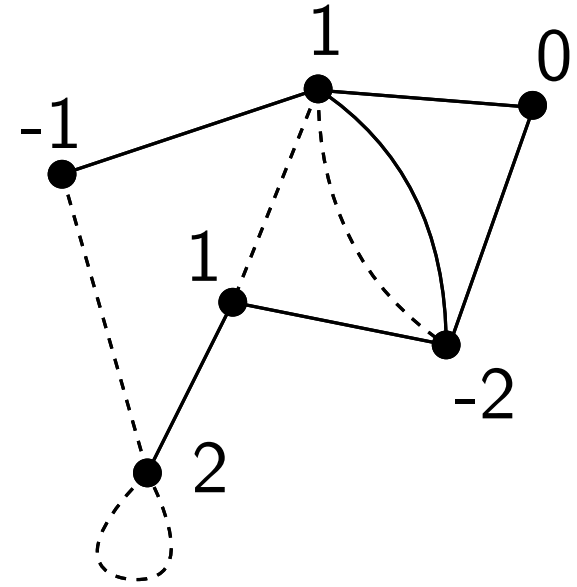
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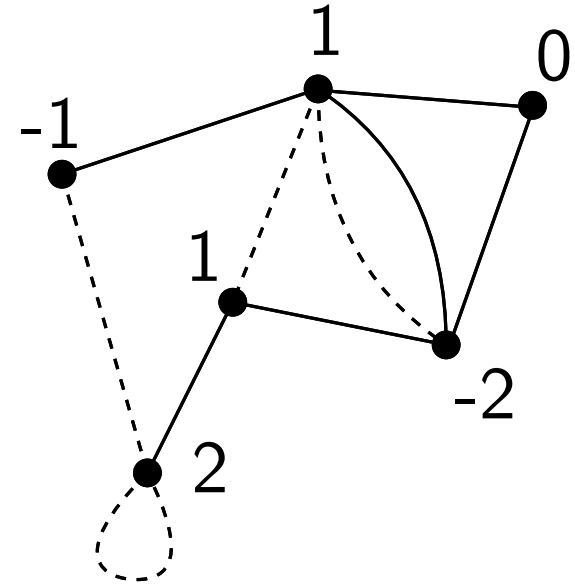


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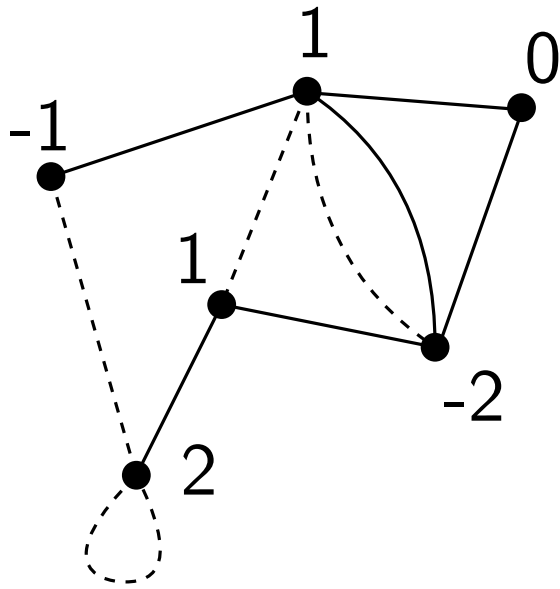
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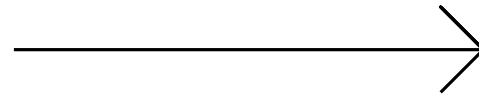
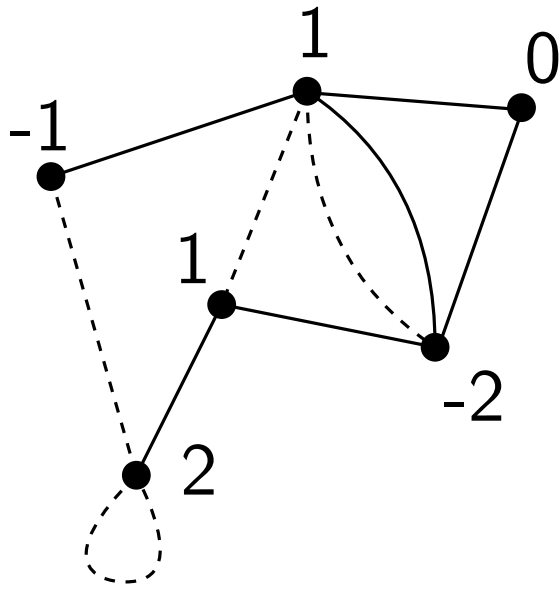
The chromatic polynomial of Σ is

$$\chi_{\Sigma}(k) := \# \left\{ \begin{array}{l} \text{proper colorings of } \Sigma \\ \text{with color set} \\ \{0, \pm 1, \dots, \pm k\} \end{array} \right\}$$

The signed-symmetric invariant



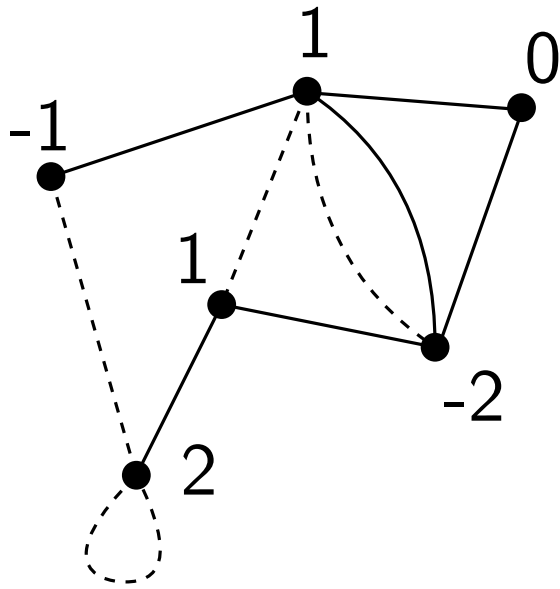
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Monomial in the set of
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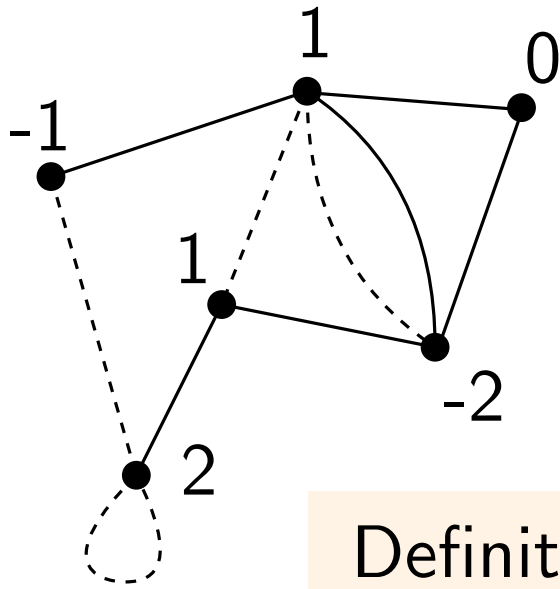


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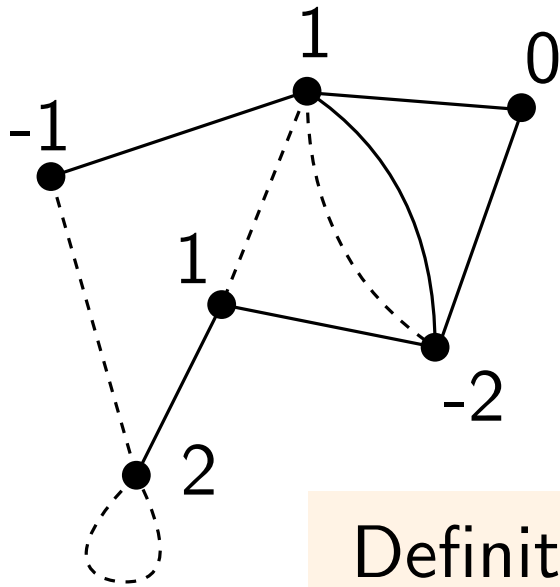
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Definition (Kuroda & Tsujie 2021)

Given a signed graph Σ we define the *signed-symmetric chromatic function* to be:

$$X_\Sigma(\dots x_{-1}, x_0, x_1, \dots) = \sum_{\kappa \text{ proper}} \mathbf{x}^\kappa$$

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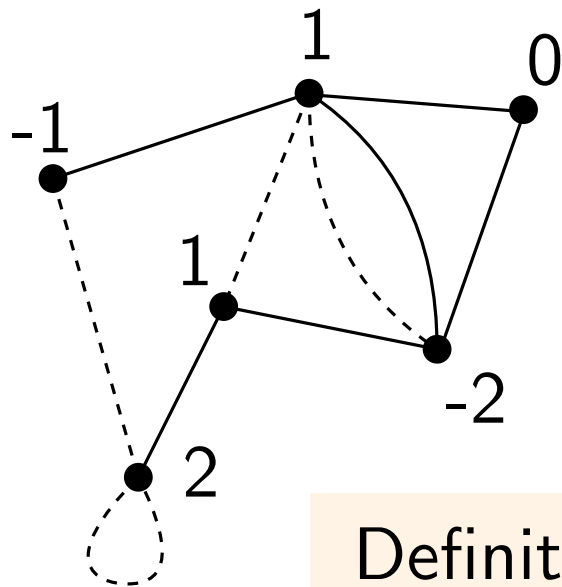
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The signed-symmetric invariant



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Example



$$X_\Sigma = 2 \sum_{i,j \in \mathbb{N}, i < j} x_i x_j + 2 \sum_{i,j \in \mathbb{N}, i \neq j} x_i x_{-j}$$

$$2 \sum_{i,j \in \mathbb{N}, i < j} x_{-i} x_{-j} + x_0 \sum_{i \in \mathbb{N}} x_i + x_0 \sum_{i \in \mathbb{N}} x_{-i}$$

Signed-symmetric functions

$$S\mathfrak{S} := \{\text{bijections } \sigma : \mathbb{Z} \rightarrow \mathbb{Z} : \sigma(i) = -\sigma(-i) \ \forall i \in \mathbb{Z}\}$$

Signed-symmetric functions

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We say a formal power series $f \in \mathbb{Q}[[\dots, x_{-1}, x_0, x_1, \dots]]$ is a *signed symmetric function* if:

- f has bounded degree
- $f(\dots, x_{-1}, x_0, x_1, \dots) = f(\dots, x_{\sigma(-1)}, x_{\sigma(0)}, x_{\sigma(1)} \dots) \ \forall \sigma \in S\mathfrak{S}$

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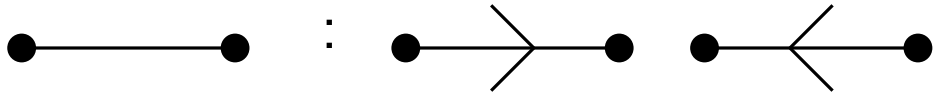
$$X_{\Sigma}(\dots, x_{-1}, x_0, x_1, \dots) \in SSym^{|V|}$$

Orientation of signed graphs

What is an orientation of a signed graph ?

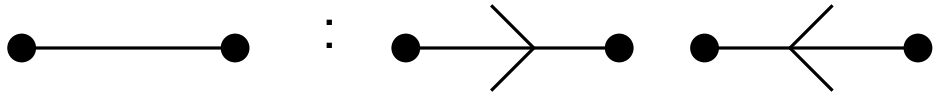
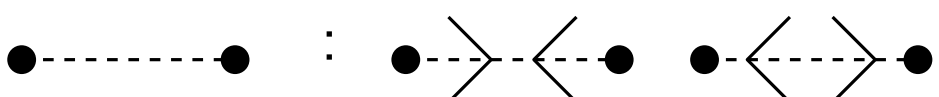
Orientation of signed graphs

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- Positive edges 

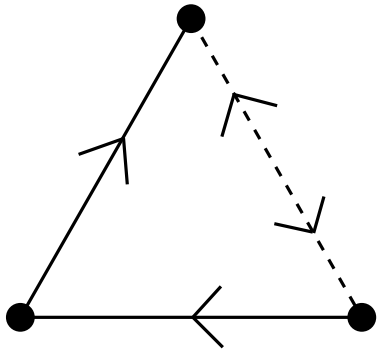
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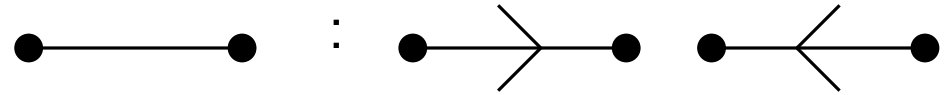
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- Negative edges 

Orientation of signed graphs

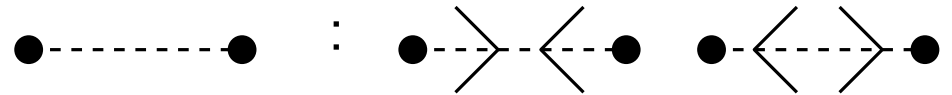
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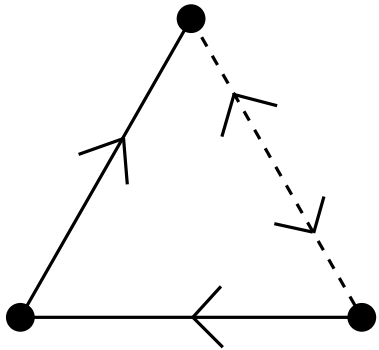


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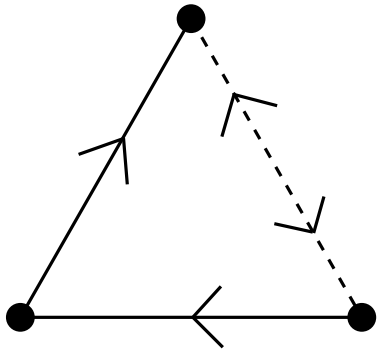


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An acyclic orientation of Σ is an orientation in which every closed walk has a source or a sink

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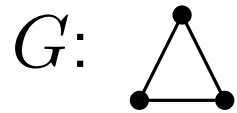
An acyclic orientation of Σ is an orientation in which every closed walk has a source or a sink

Theorem (Zaslavsky):

$$\chi_{\Sigma}(-1) = \#\{\text{acyclic orientations of } \Sigma\}$$

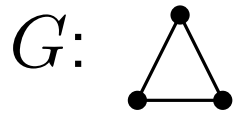
Summary

simple graph



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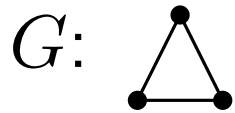


Stanley 1995

$$\longrightarrow X_G(x_1, x_2, \dots) \in Sym$$

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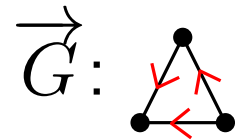
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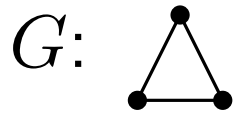


Ellzey 2017

$$\longrightarrow X_{\vec{G}}(x_1, x_2, \dots; \textcolor{red}{t}) \in QSym[t]$$

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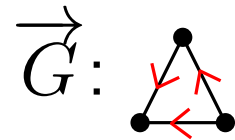
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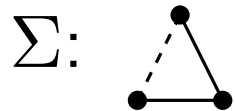
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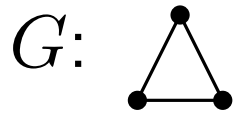
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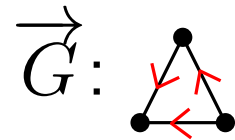
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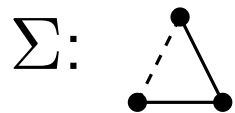
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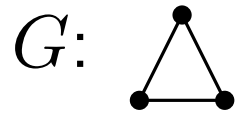


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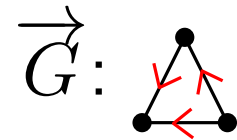
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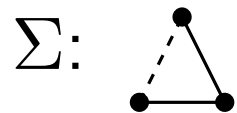
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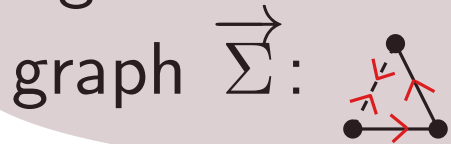
signed graph



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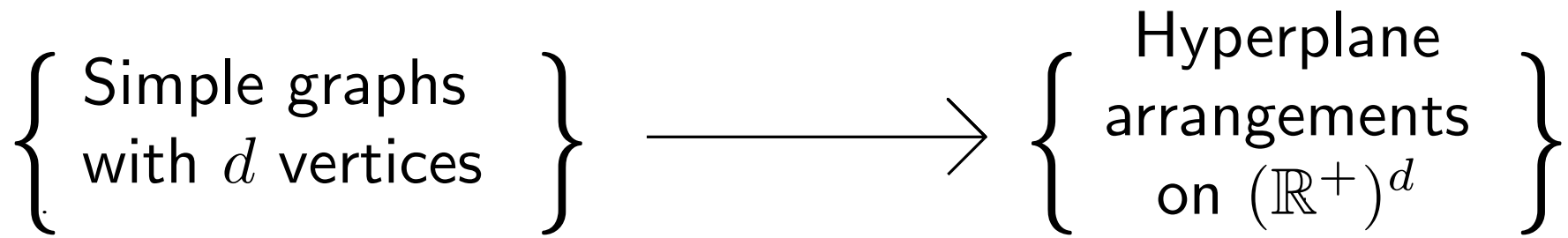


$$\longrightarrow X_{\vec{\Sigma}}(\dots x_{-1}, x_0, x_1, \dots; \textcolor{red}{t}) \in \textcolor{red}{?}[t]$$

II. How to define

$$X_{\overrightarrow{\Sigma}}(\dots x_{-1}, x_0, x_1, \dots; t)?$$

The geometric viewpoint



The geometric viewpoint

$$\left\{ \begin{array}{l} \text{Simple graphs} \\ \text{with } d \text{ vertices} \end{array} \right\} \longrightarrow \left\{ \begin{array}{l} \text{Hyperplane} \\ \text{arrangements} \\ \text{on } (\mathbb{R}^+)^d \end{array} \right\}$$

$$G = (V, E) \longmapsto \mathcal{H}_G = \{x_i = x_j : \{i, j\} \in E\}$$

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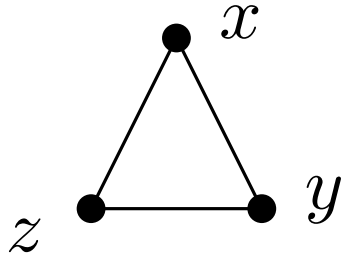
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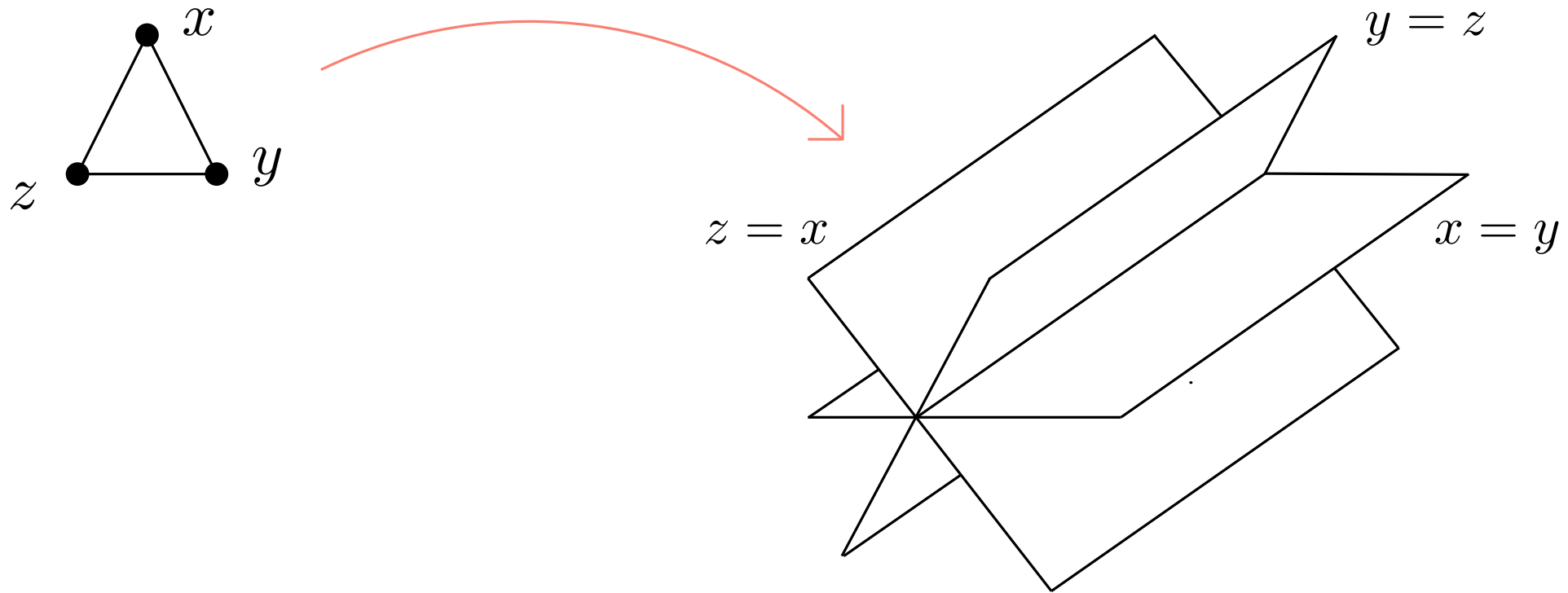
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$$\left\{ \begin{array}{l} \text{Acyclic orientations} \\ \text{of } G \end{array} \right\} \xleftrightarrow{\text{Greene}} \left\{ \begin{array}{l} \text{Connected} \\ \text{components of} \\ (\mathbb{R}^+)^d \setminus \mathcal{H}_G \end{array} \right\}$$

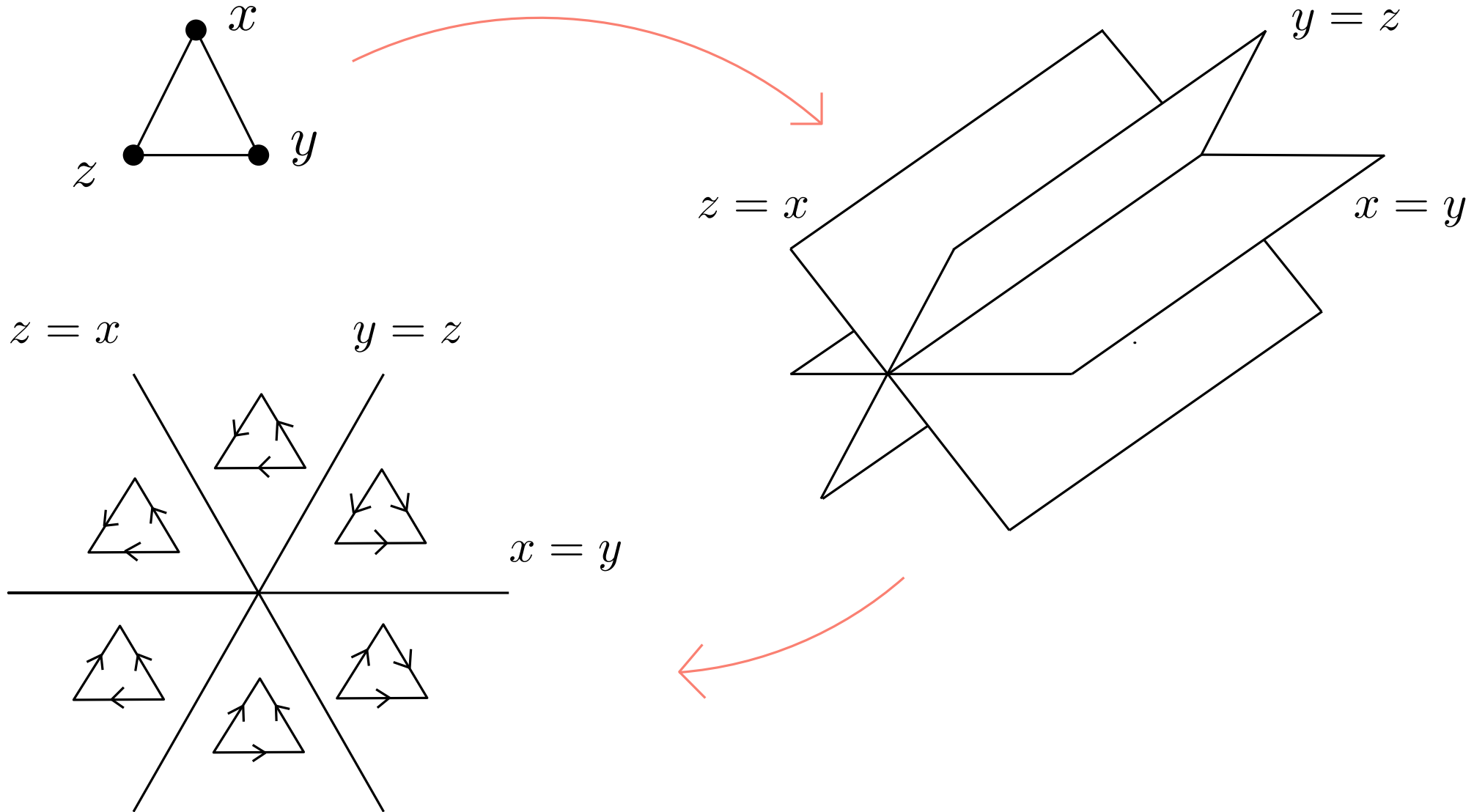
The geometric viewpoint: an example



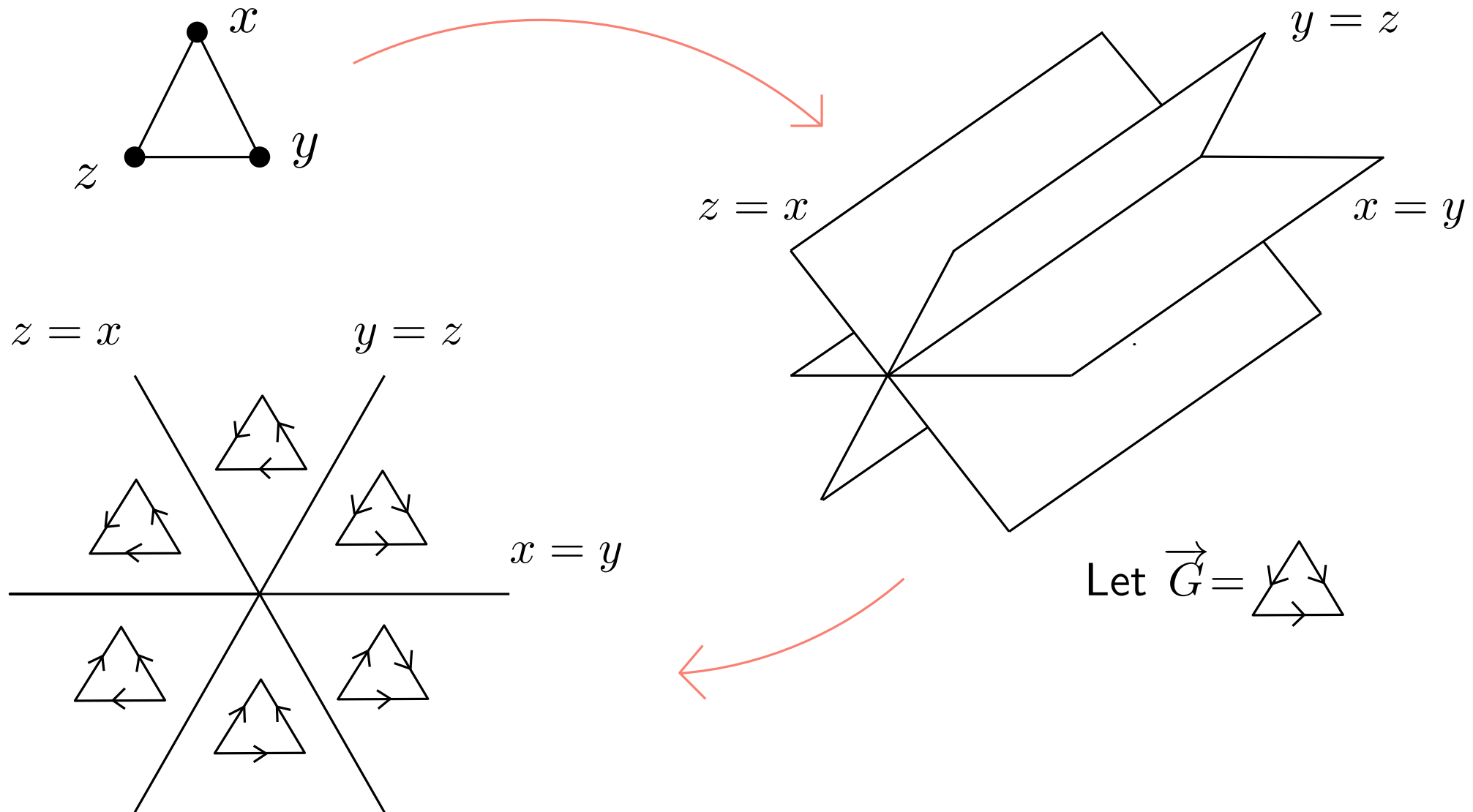
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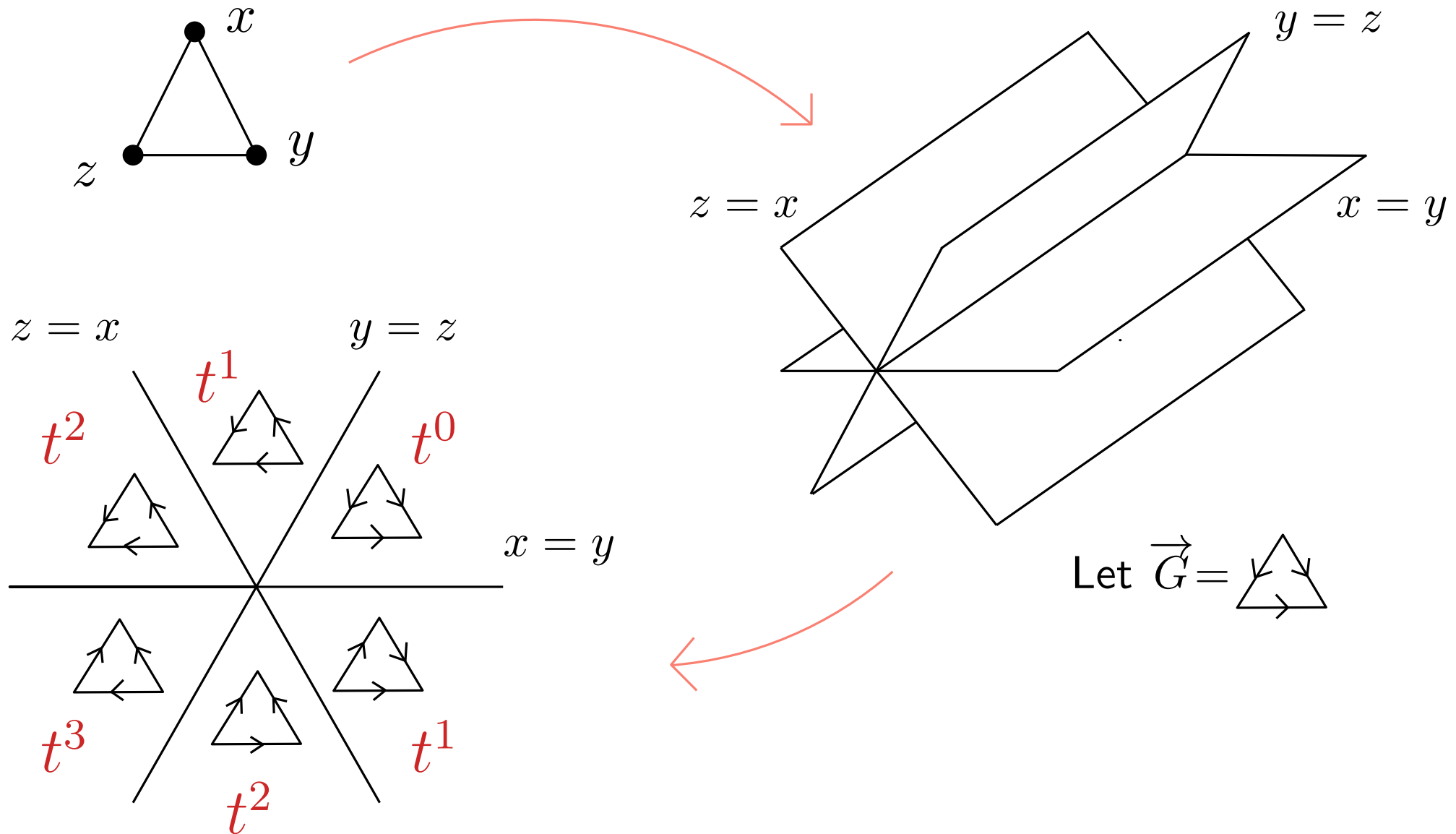
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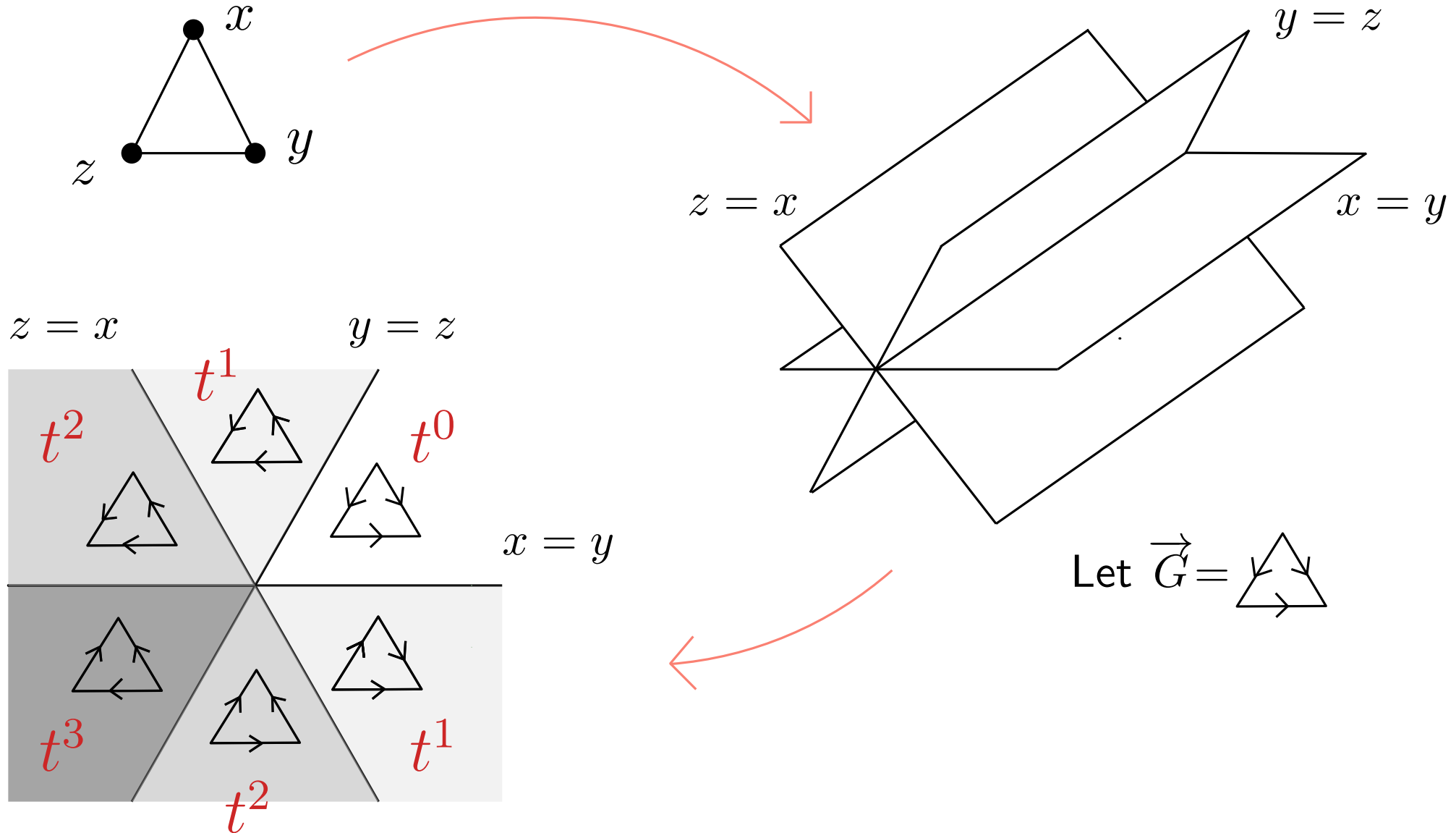
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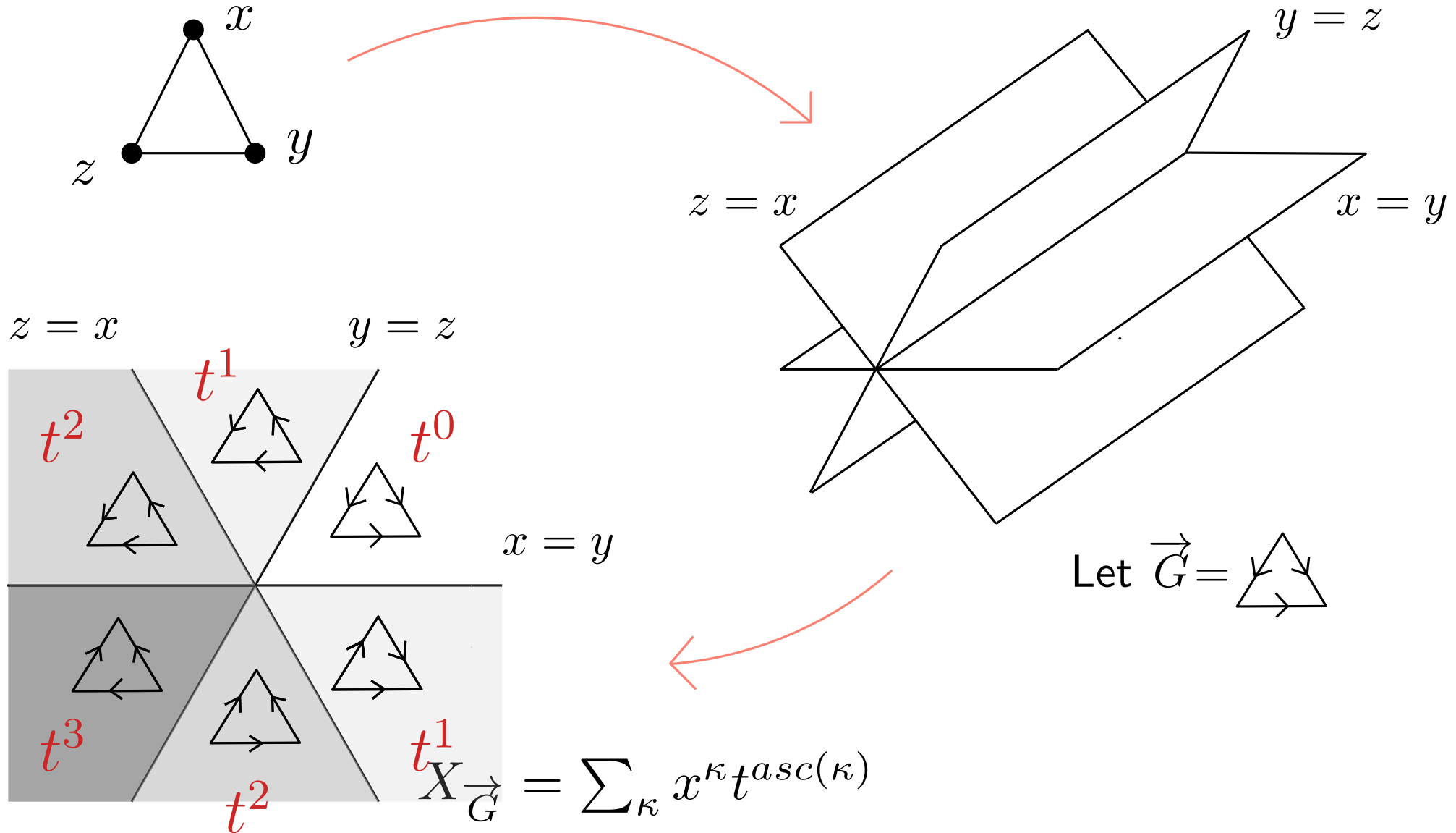
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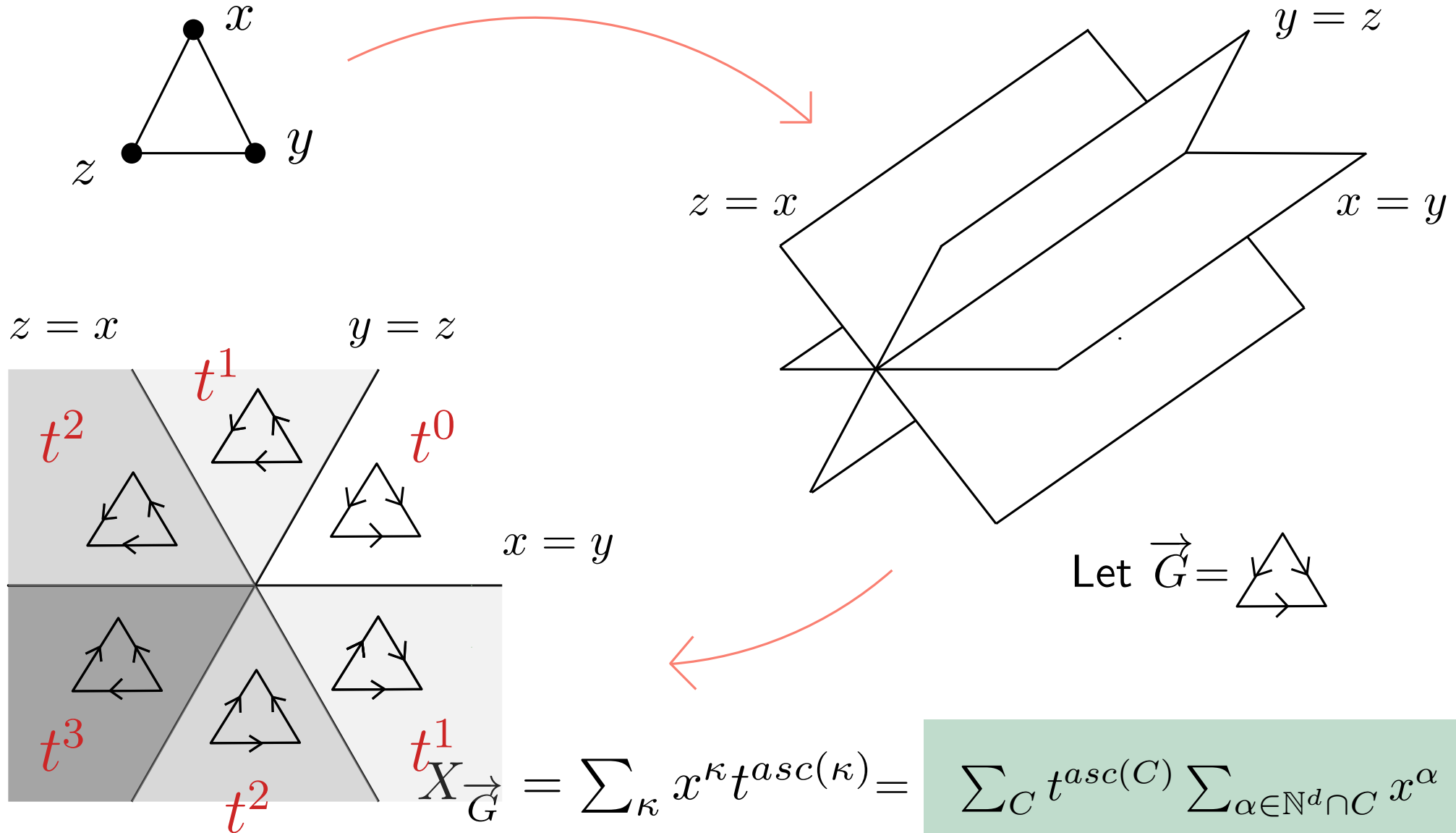
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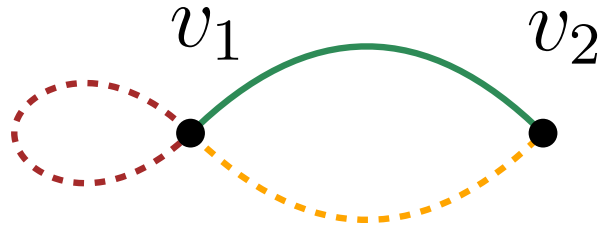
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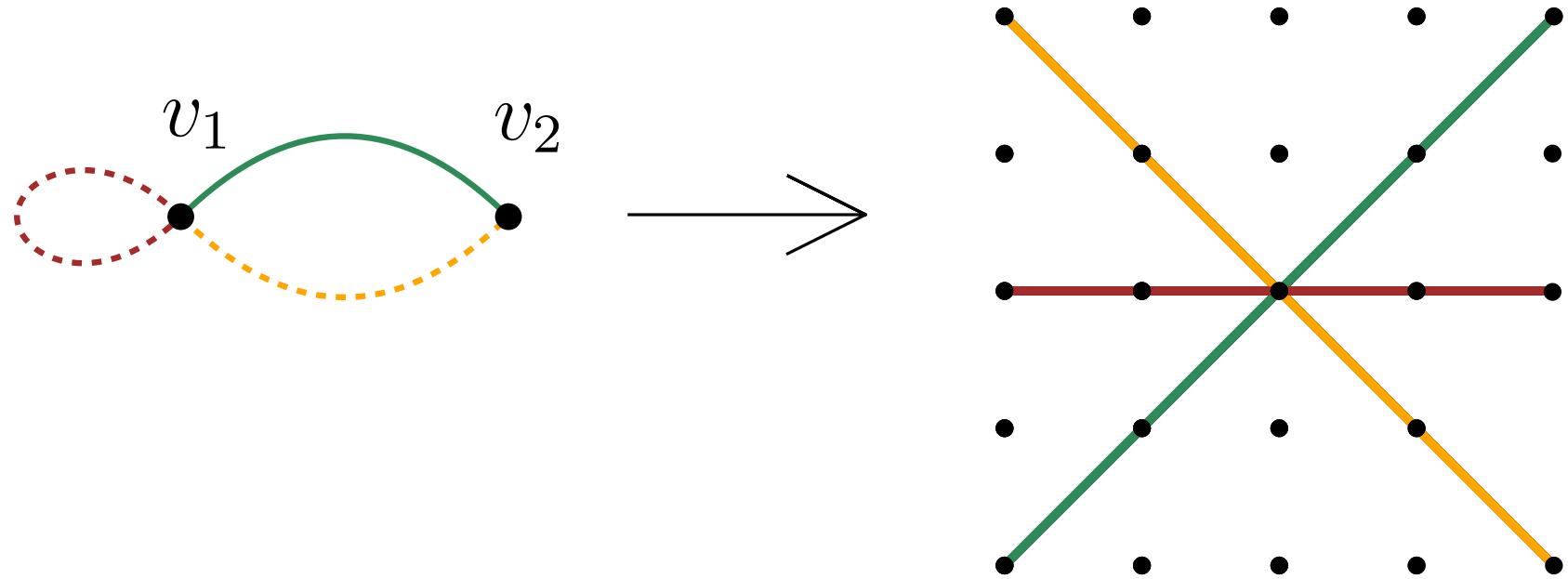
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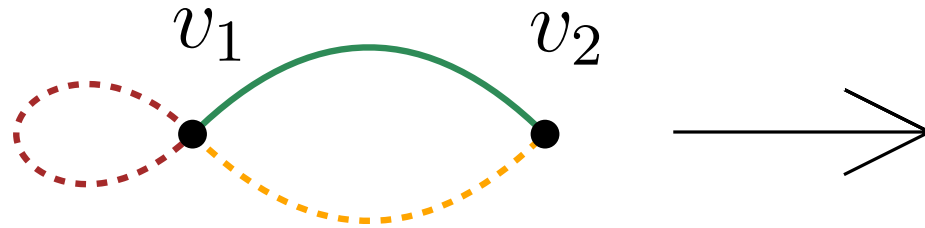
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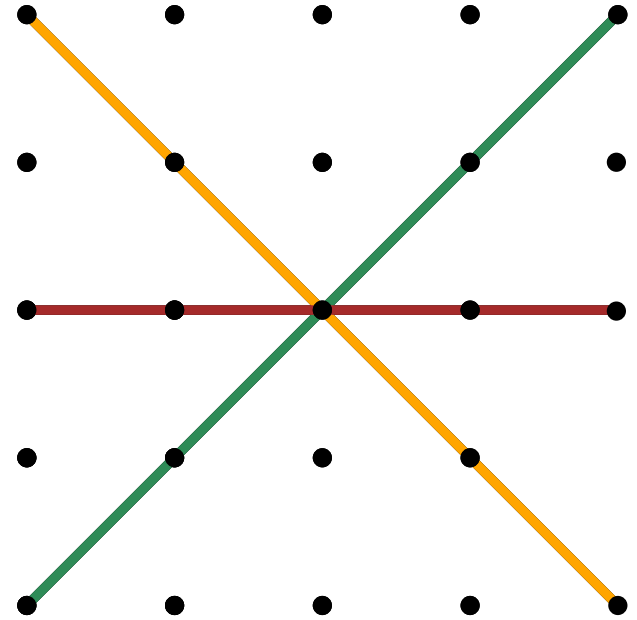


The geometric viewpoint: signed case

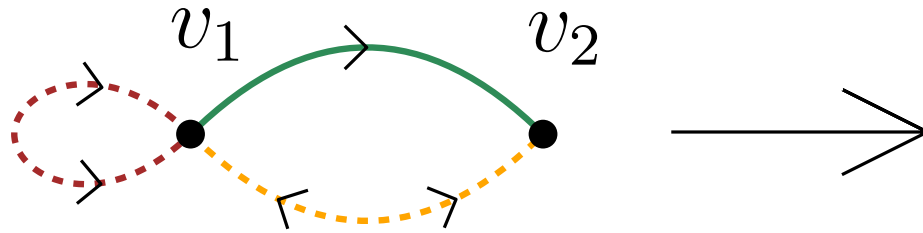


Theorem (Zaslavsky '83):

There is a bijection between acyclic orientations of Σ and the chambers of $\mathcal{H}_\Sigma$

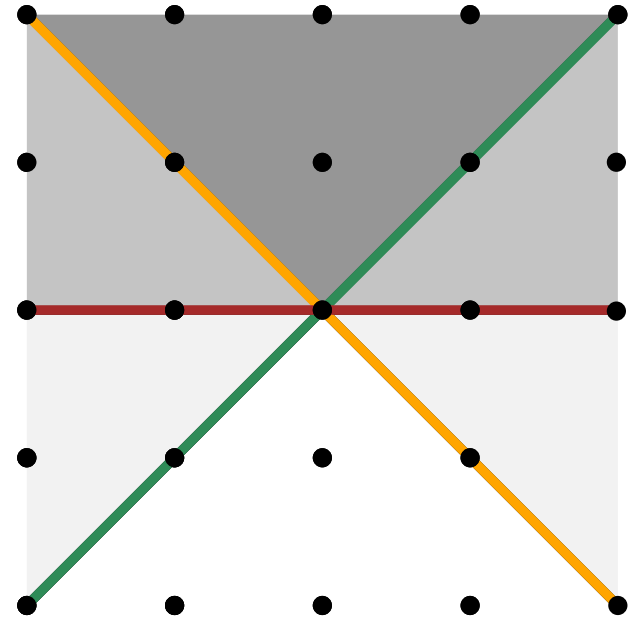


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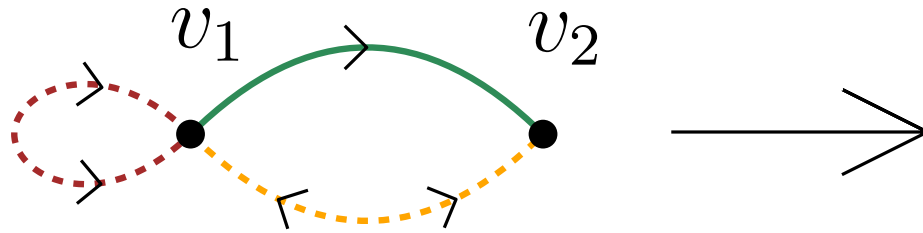


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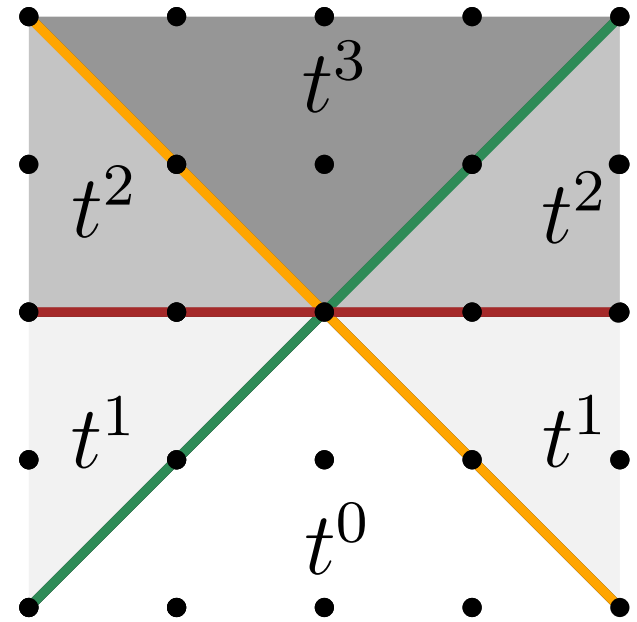


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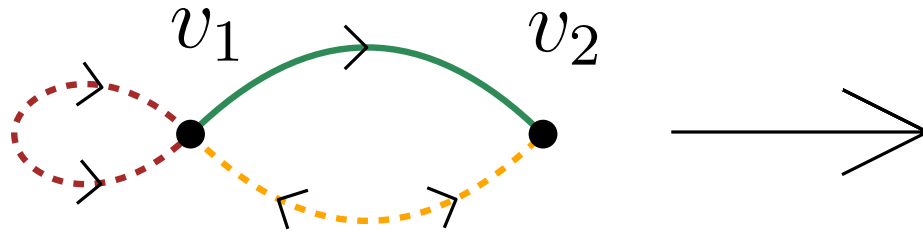


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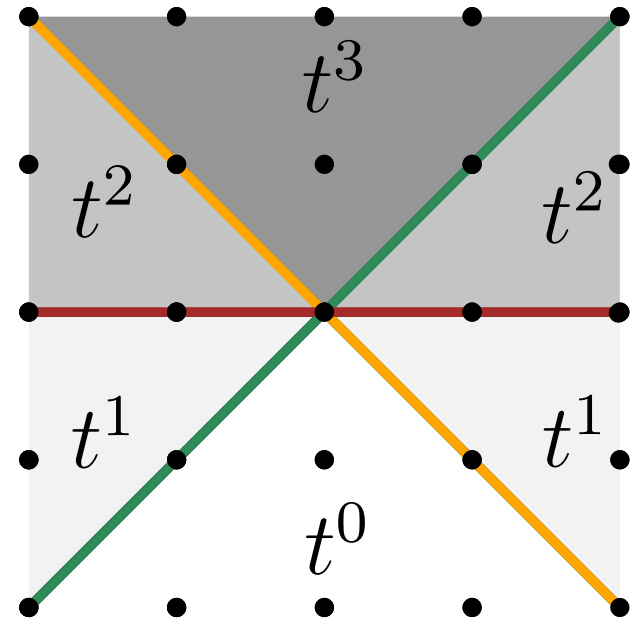


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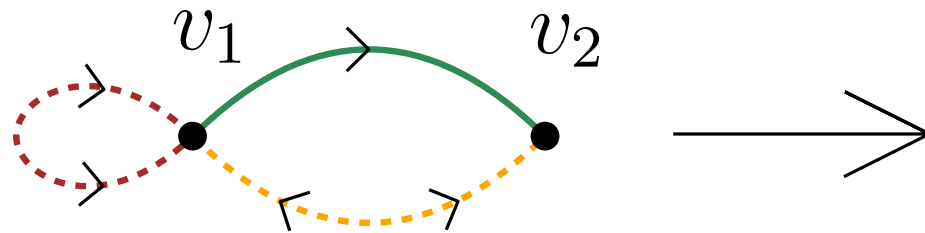
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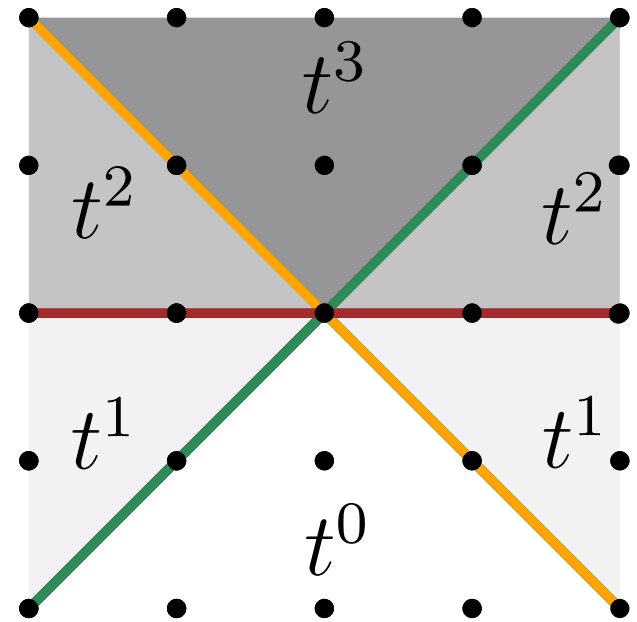
$$X_{\vec{\Sigma}}(\mathbf{x}; t) := \sum_C t^{\text{asc}(C)} \sum_{\alpha \in \mathbb{Z}^d \cap C} \mathbf{x}^\alpha$$

The geometric viewpoint: signed case



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$$X_{\vec{\Sigma}}(\mathbf{x}; t) := \sum_C t^{\text{asc}(C)} \sum_{\alpha \in \mathbb{Z}^d \cap C} \mathbf{x}^\alpha = \sum_{\kappa} t^{\text{asc}(\kappa)} \mathbf{x}^\kappa$$

The definition

$$X_{\vec{\Sigma}}(\dots, x_{-1}, x_0, x_1 \dots; t) = \sum_{\kappa \text{ proper}} x^{\kappa} t^{asc(\kappa)}$$

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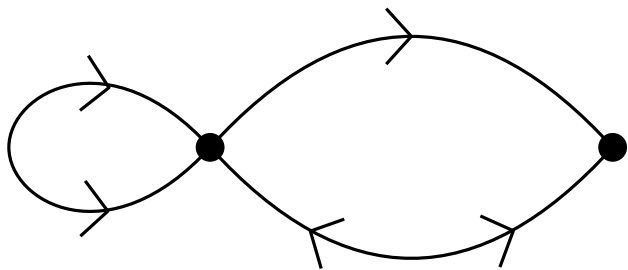
$$\left\{ \begin{array}{c} \text{positive edges} \\ \bullet \xrightarrow{i} \bullet \text{ verifying} \\ \kappa(i) < \kappa(j) \end{array} \right\} \cup \left\{ \begin{array}{c} \text{introverted edges} \\ \bullet \xrightarrow{i} \bullet \xleftarrow{j} \bullet \text{ verifying} \\ \kappa(i) + \kappa(j) < 0 \end{array} \right\} \cup \left\{ \begin{array}{c} \text{extroverted edges} \\ \bullet \xleftarrow{i} \bullet \xrightarrow{j} \bullet \text{ verifying} \\ \kappa(i) + \kappa(j) > 0 \end{array} \right\}$$

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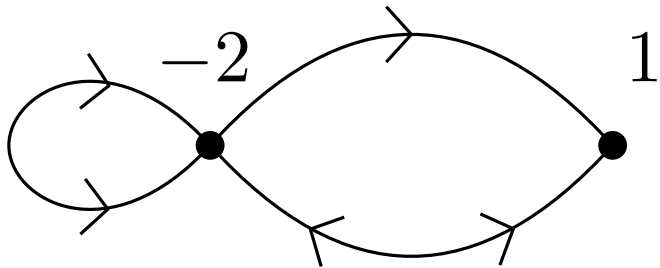


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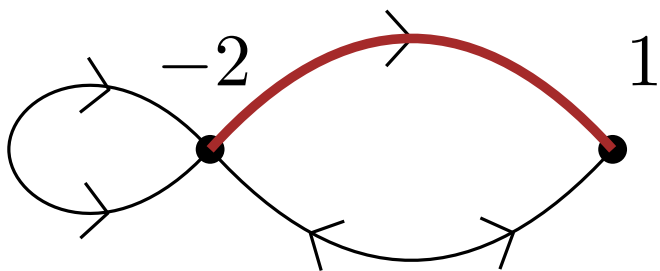


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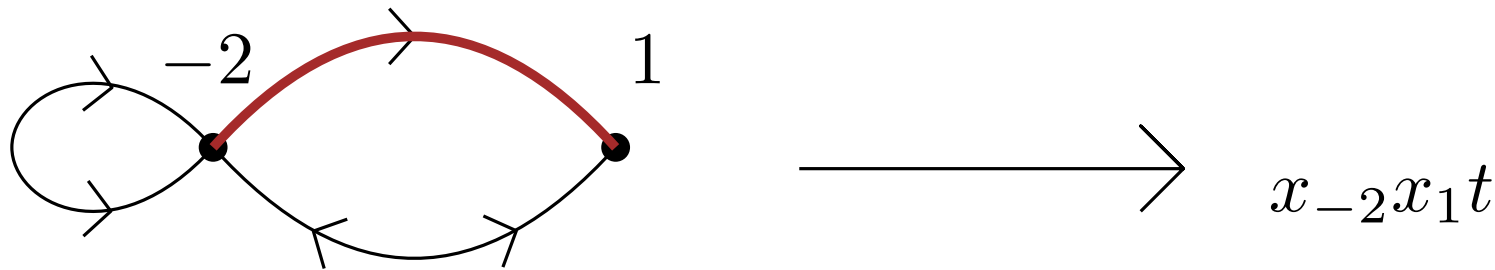


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III. The algebra
 $SQSym$

The definition of the algebra

The bases of $SQSym$ are indexed by pairs

$(k, \lambda) = (k, \begin{pmatrix} a_1 & a_2 & \cdots & a_r \\ b_1 & b_2 & \cdots & b_r \end{pmatrix})$ where $k \in \mathbb{N}_0$ and $(a_i, b_i) \neq (0, 0)$

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The monomial basis $\{M_{(k, \lambda)}\}_{(k, \lambda)}$ is given by

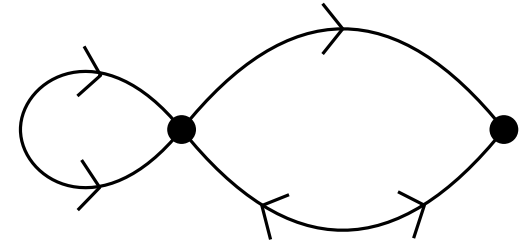
$$M_{(k, \lambda)} := \sum_{i_1 < \cdots < i_r} x_0^k x_{i_1}^{a_1} x_{-i_1}^{b_1} \cdots x_{i_r}^{a_r} x_{-i_r}^{b_r}$$

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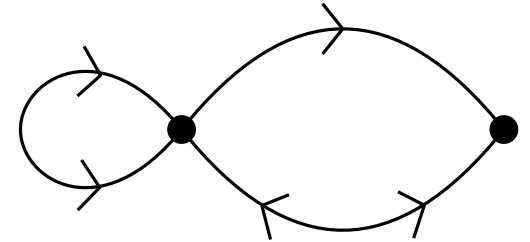


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$$\begin{aligned} X_{\vec{\Sigma}}(\mathbf{x}; t) = & t^3 [M_{0, \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}}] + t^2 [2M_{0, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}} + M_{0, \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}} + M_{1, \begin{pmatrix} 1 \\ 0 \end{pmatrix}}] + \\ & t [2M_{0, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}} + M_{0, \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}} + M_{1, \begin{pmatrix} 0 \\ 1 \end{pmatrix}}] + [M_{0, \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}}] \end{aligned}$$

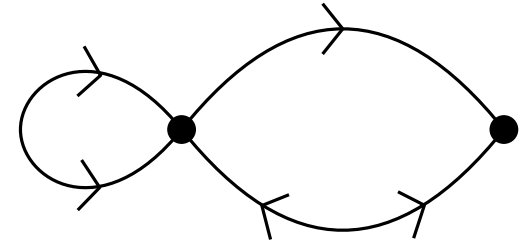
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We can define $SQSym := \text{span}_{\mathbb{Q}}\{M_{(k, \lambda)}\}$

Sequence of dimensions

If $SQSym^d$ are the homogeneous signed symmetric functions of degree d then

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Its generating function is

$$\sum_{d \geq 0} \dim(SQSym^d) t^d = \frac{1-t}{1-4t+2t^2}$$

The fundamental generating family

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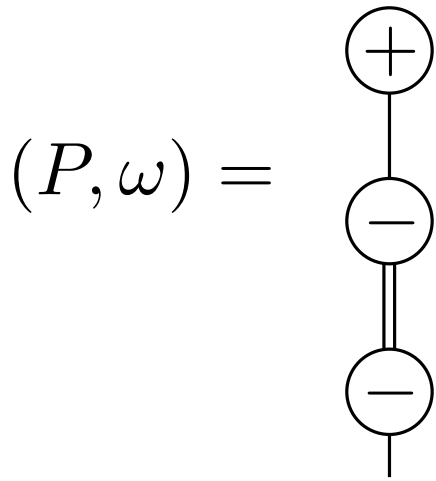
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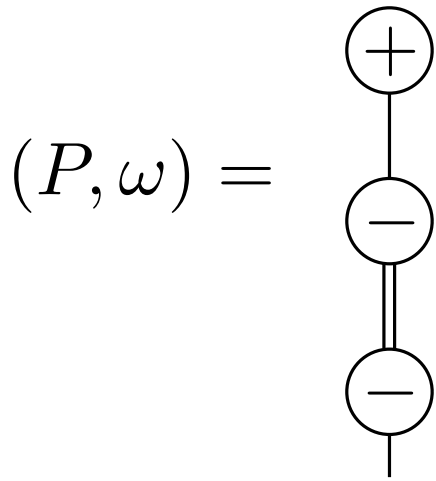
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But we know how to extract a basis! $\overline{\mathcal{F}} \subset \mathcal{F}$

A formula for $X_{\overrightarrow{\Sigma}}(\mathbf{x}; t)$

A formula for $X_{\vec{\Sigma}}(\mathbf{x}; t)$

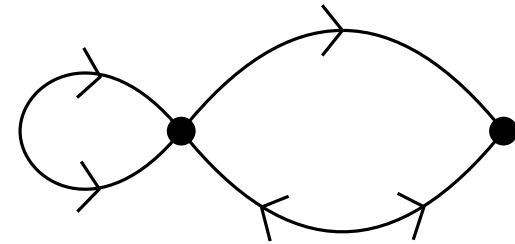
Theorem: Given a directed signed graph $\vec{\Sigma}$ on d vertices then

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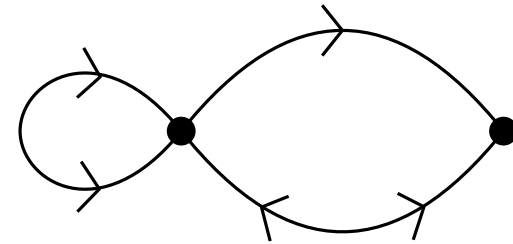


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$\overline{2}112$	3	++	1234	$\{0,1\}$
$\overline{1}221$	2	++	1223	$\{0,1\}$
$2\overline{1}1\overline{2}$	1	+-	1234	$\{0,1\}$
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$1\overline{2}2\overline{1}$	1	+-	1223	$\{0,1\}$
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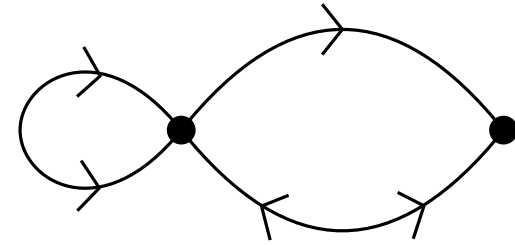


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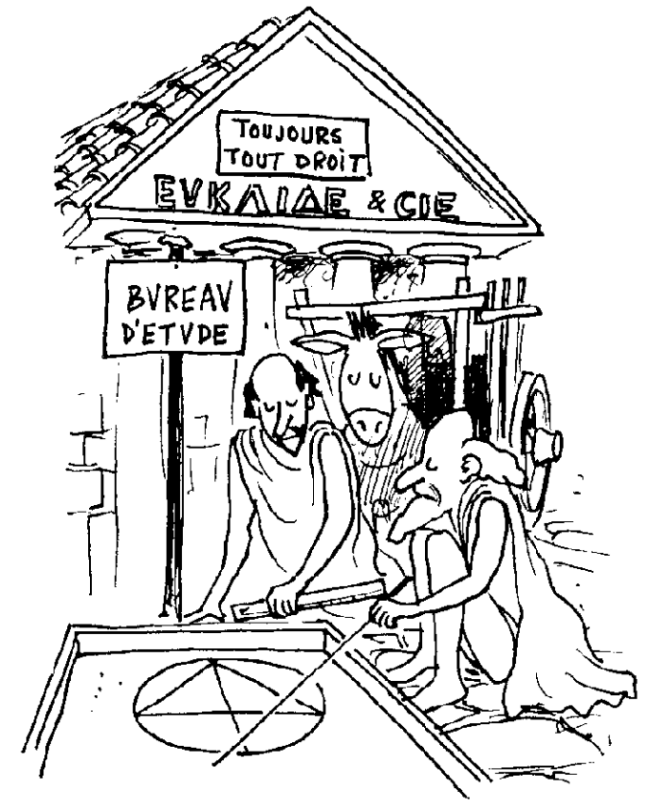


$$\begin{aligned} X_{\vec{\Sigma}}(\mathbf{x}; t) = & t^3 \Gamma_{\{0,1\}}^{++} + \\ & t^2 [\Gamma_{\{0,1\}}^{++} + \Gamma_{\{0,1\}}^{-+} + \Gamma_{\{1\}}^{-+}] + \\ & t [2\Gamma_{\{0,1\}}^{+-} + \Gamma_{\{1\}}^{--}] + \Gamma_{\{0,1\}}^{--} \end{aligned}$$

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